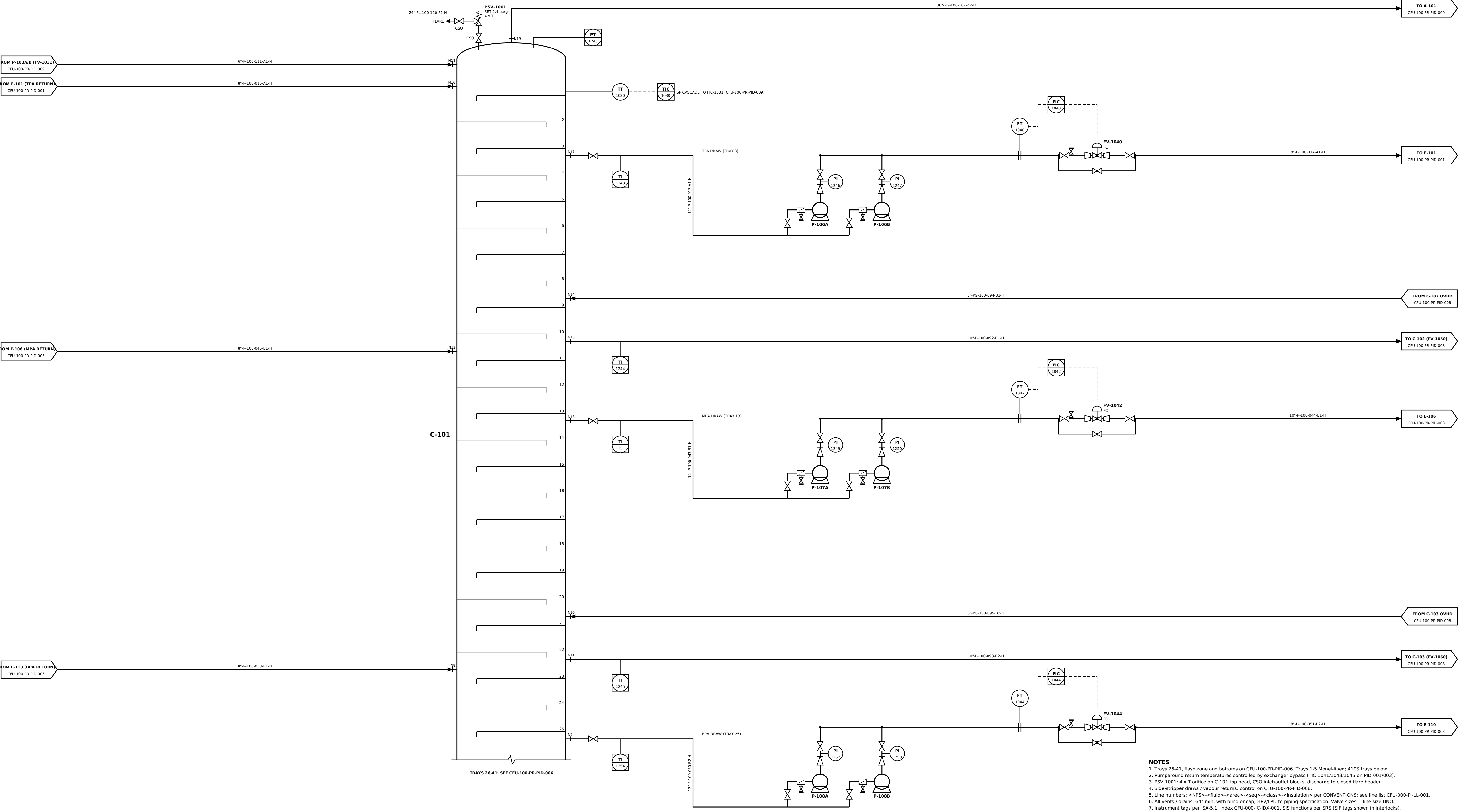


C-101
Atmospheric crude fractionator
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T
DESIGN 3.5 barg / 390 C; CA 3 mm
41 valve trays (2/4-pass), 4105; Monel-lined top 5 trays
MAT'L CS + 4105 clad below tray 10; Monel 400 lined top head & trays
1-5

P-106A/B
TPA pumparound
RATED 293 m³/h @ 130 m, dP 8 bar
MOTOR 1104 kW, API 682 Plus 1152
DES. T 185 C; MAT'L S-6 (CS/12Cr)

P-107A/B
MPA pumparound
RATED 387 m³/h @ 138 m, dP 8 bar
MOTOR 1364 kW, API 682 Plus 1152
DES. T 270 C; MAT'L C-6 (12Cr)

P-108A/B
BPA pumparound
RATED 318 m³/h @ 142 m, dP 8 bar
MOTOR 1324 kW, API 682 Plus 538
DES. T 345 C; MAT'L A-8 (316)



- NOTES**
- Trays 26-41, flash zone and bottoms on CFU-100-PR-PID-006. Trays 1-5 Monel-lined; 4105 trays below.
 - Pumparound return temperatures controlled by exchanger bypass (TIC-1041/1043/1045 on PID-001/003).
 - PSV-1001: 4 x T orifice on C-101 top head. CSO inlet/outlet blocks: discharge to closed flare header.
 - Side-stripper draws / vapour returns: control on CFU-100-PR-PID-008.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	8 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-007			A